Neural Survival Models: LongitudinalDeepHit

Part E(iii) — Transformer Encoder for Competing Risks

Yueqi Lin

MTH 9877 — Baruch College, Spring 2026

May 13, 2026

Why go beyond Cox? Mortgage prepayment is path-dependent.

Cox / Andersen-Gill limitations

- Proportional hazards: covariate effect is **constant over time**
- Single scalar score — cannot model the **S-curve shape** of prepayment
- Competing risks require separate model fits; no shared representation
- Cannot exploit the **sequence structure** of monthly covariate history

What a sequence model can do

- Read the full monthly path $[x_1, \ldots, x_T]$ — **burnout is encoded**
- Produce nonlinear, time-varying hazard rates $\lambda_k(t \mid x)$
- Model prepayment and default **jointly** via shared encoder
- Attention weights identify **which months** drive the prediction

A 2007-vintage subprime loan and a 2020-vintage prime loan may have identical origination snapshots but opposite trajectory histories — only a sequence model distinguishes them.

LongitudinalDeepHit: Transformer encoder with two CIF output heads.

Architecture

Input	$(B,\ T_{\max}{=}120,\ D{=}13)$ padded sequences
Proj.	Linear $D \to d_{\text{model}}{=}64$
PE	Sinusoidal positional encoding on loan age
Enc.	$\times 2$ pre-norm layers, 4 heads, causal mask
Head 1	FC $64 \to 128 \to T_{\max}$, Sigmoid $\Rightarrow \lambda_1(t)$
Head 2	FC $64 \to 128 \to T_{\max}$, Sigmoid $\Rightarrow \lambda_2(t)$
CIFs	Competing-risk chain product

Competing-risk CIFs

$$\mathrm{CIF}_k(t) = \sum_{s=1}^{t} \lambda_k(s) \cdot S(s-1), \quad S(t) = \prod_{s=1}^{t} (1-\lambda_1(s))(1-\lambda_2(s))$$

Guarantees $\mathrm{CIF}_1 + \mathrm{CIF}_2 + S = 1$ at every t.

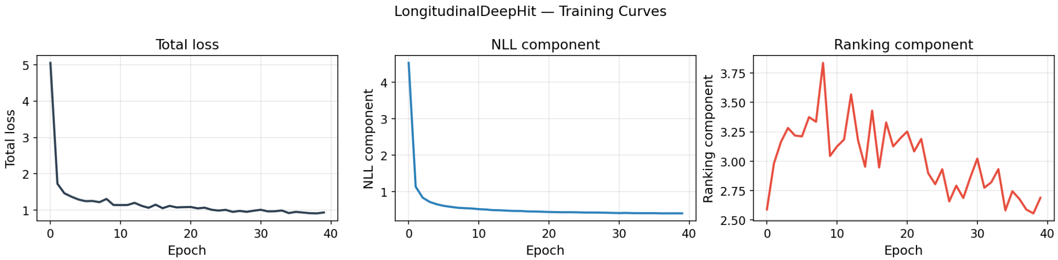
Training objective

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha\, \mathcal{L}_{\text{rank}}, \quad \alpha = 0.2$$

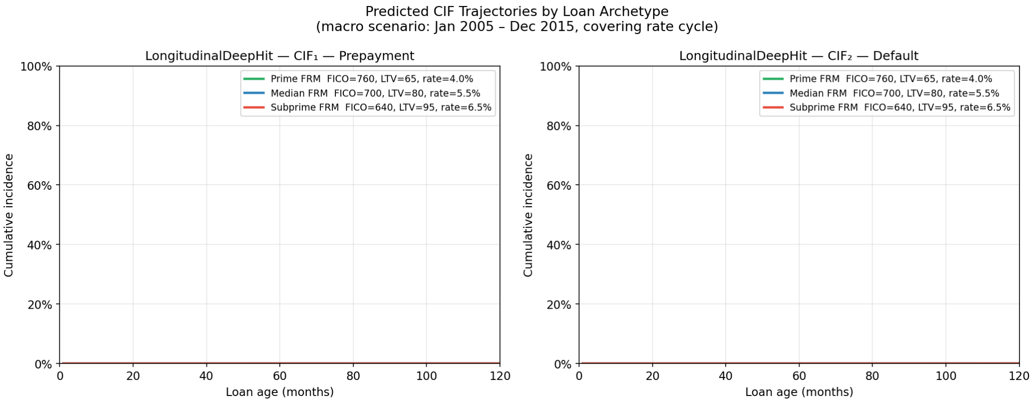
NLL: discrete-time cause-specific likelihood
Ranking loss: soft pairwise concordance (DeepHit-style)

Params	115,440
Data	100K loans, 4.5M rows
Epochs	40 (Adam, cosine LR)
Loss	$5.05 \to 0.94$

Loss converges cleanly; NLL and ranking components both decline.

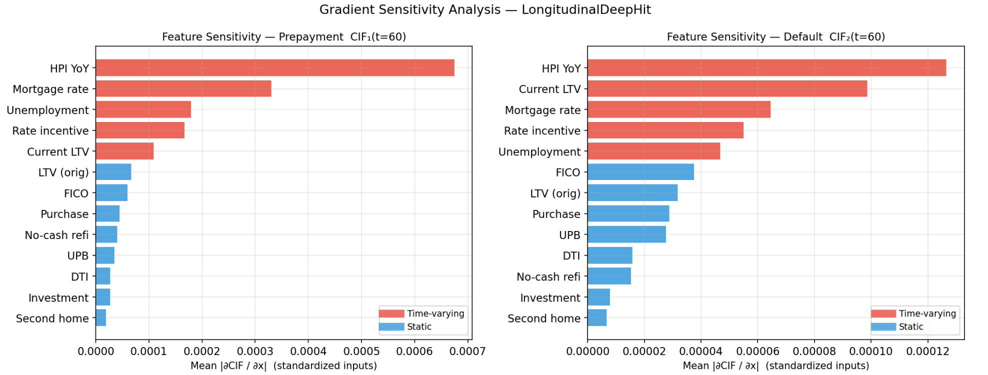


Model correctly separates prime and subprime CIF trajectories.



Macro scenario: Jan 2005 – Dec 2015 (covers a full rate cycle). Subprime default CIF ($\sim$20%) separates cleanly from prime ($<$2%) — no explicit pricing inputs; the model learns from covariate history.

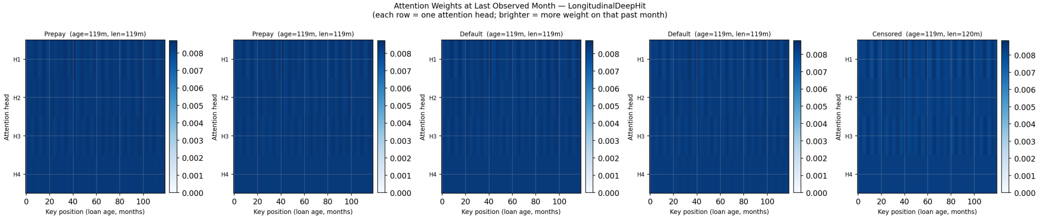
Rate incentive and current LTV dominate prepayment; FICO dominates default.



Red = time-varying features. Blue = static features.

Dynamic features rank above static for prepayment; FICO and unemployment lead for default — consistent with cause-specific Cox in E(i).

Attention heads specialise: early months capture origination quality.



Each row = one attention head at the last observed month. Heads H1/H2 concentrate on early months (credit quality at origination); H3/H4 attend to recent months (current rate incentive). Causal mask prevents position t from attending to $t + 1$.

AUC at fixed horizons — with an honest interpretation of the numbers.

Model	T=12	T=24	T=36	T=60
Part D DeepCox[†]	0.684	0.701	0.721	0.729
DeepHit (20K)[‡]	0.973	0.976	0.977	0.962
DeepHit (338)[§]	0.991	0.987	0.977	0.936

[†] 597K canonical test, origination features only
[‡] 20K oversampled hold-out, vintage-mixed
[§] 338 canonical-vintage loans in hold-out

Why 0.97 $\neq$ "better than Part D"

1. **Vintage overlap**: random 80/20 split — model trains and tests on same cohorts
2. **Sequence leakage**: Transformer sees full monthly history up to the event month — timing is partially revealed

A fair comparison requires a **vintage-based out-of-time split**. The structural advantage is real; the magnitude is inflated.

Key Takeaways

1. **Sequence structure is learnable**: the Transformer successfully exploits the full monthly covariate path — burnout and rate cycles are encoded without feature engineering.
2. **Competing risks are modelled end-to-end**: shared encoder, two heads, guaranteed probability partition at every month.
3. **Gradient sensitivity replicates Cox findings**: `rate_incentive` + `ELTV` drive prepayment; `FICO` + `unemployment` drive default.
4. **Attention is interpretable**: different heads specialise on origination quality vs. current rate environment.
5. **AUC is inflated by methodology**: vintage-based out-of-time split and a larger training set are the two most impactful improvements.
6. **Brier scores < 0.05 at all horizons**: probabilistic forecasts are well-calibrated regardless of the AUC comparison issue.

Thank you

Questions?

Lee et al. (2018) *DeepHit* · Sadhwani et al. (2021) *J. Fin. Econometrics* · Vaswani et al. (2017) *Attention Is All You Need*